

Question

A is a metallic element, and its powder is grayish-black. The element **A** reacts with fluorine gas upon heating to generate a fluoride molecule **B** with an octahedral configuration. **B** disproportionates in water to produce **C** and **D**. **C** is a dark brown solid with a rutile structure; **D** is a diamagnetic light yellow solid with a low melting point and is easily volatile. **C** can also be obtained by heating in air. In potassium hydroxide solution, **D** reacts with an equimolar amount of potassium nitrite to produce a 2:1 type potassium salt **E**; **E** can be heated in nitric acid to obtain **D**. **B** reacts with an equimolar amount of **D** at 150°C to produce a compound **F** (containing **A** 68.1%). Structural determination shows that **F** is a polymer with two chemical environments of **A** atoms, **A1** and **A2** (both octahedrally coordinated), and the bond lengths of the chemical bonds they form are as follows:

A1: 170.7pm, 182.0pm, 182.8pm, 185.5pm, 202.9pm, 204.4pm

A2: 170.6pm, 171.1pm, 171.1pm, 183.4pm, 214.0pm, 214.2pm

Which of the following statements is correct:

- A.** All other options are incorrect
- B.** The atomic ground state of element **A** has four unpaired electrons.
- C.** **D** is non-toxic
- D.** The chemical equation that generates **E**, when the coefficients are balanced to the simplest integer ratio, has a sum of coefficients of all species greater than 10.
- E.** The chemical equation for **E** heated in nitric acid, when the coefficients are balanced to the simplest integer ratio, the sum of the reactant coefficients and the sum of the product coefficients are both not prime numbers.

F. **B** molecule has an octahedral configuration.

G. According to the information in the question, the magnetic moment of the **F** molecule can be predicted to be almost 0.

Answer

Correct Answer: **G**

Detailed Explanation

According to the information in the question, it is obvious that **B** is a hexafluoride, **C** is a dioxide, and **D** is a highest-valence oxide.

CHECKPOINT

1 PTS

B is a hexafluoride, **C** is a dioxide, and **D** is a highest-valence oxide

B, **D** react in equimolar amounts to obtain **F**, indicating that **F** contains F, O elements. Let the chemical formula of **F** be AF_xO_y , and try different values for x and y:

CHECKPOINT

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F contains F, O elements

$x=y=1$, $\text{Mr}(\text{A}) = 74.7$, no reasonable metal.

$x=1, y=2$, $\text{Mr}(\text{A}) = 108.9$, no reasonable metal.

$x=y=2$, $\text{Mr}(\text{A}) = 149.4$, no reasonable metal.

$x=2, y=1$, $\text{Mr}(\text{A}) = 115.3$, no reasonable metal.

$x=3, y=2$, $M_r(\mathbf{A}) = 190.0$, which is Os.

The others have no reasonable solutions.

Therefore, element **A** is Os. Then, **B** is OsF_6 , **C** is OsO_2 , **D** is OsO_4 , **F** is OsO_2F_3 .

CHECKPOINT

2 PTS

Based on the mass fraction deduction, element **A** is Os

CHECKPOINT

1 PTS

B is OsF_6

CHECKPOINT

1 PTS

D is OsO_4

CHECKPOINT

1 PTS

C is OsO_2

CHECKPOINT

1 PTS

F is OsO_2F_3

Based on the properties, options B and C are both incorrect.

OsO_4 reacts with equimolar KNO_2 in potassium hydroxide to produce a 2:1 type potassium salt. It is easy to think that this reaction is a redox reaction. Since it is an equimolar reaction, nitrite is oxidized to nitrate, so Os(VIII) is reduced to Os(VI) ; the common divalent anion of Os(VI) under alkaline conditions is $\text{OsO}_2(\text{OH})_4^{2-}$. Therefore, **E** is $\text{K}_2[\text{OsO}_2(\text{OH})_4]$. Therefore, balance the chemical equation of the reaction:

CHECKPOINT

1 PTS

Nitrite is oxidized to nitrate, so Os(VIII) is reduced to Os(VI)

CHECKPOINT

1 PTS

E is $\text{K}_2[\text{OsO}_2(\text{OH})_4]$



CHECKPOINT

1 PTS



The sum of the coefficients is 7, so option D is incorrect.

$\text{K}_2[\text{OsO}_2(\text{OH})_4]$ releases NO in nitric acid, producing OsO_4 . Balance the redox equation:



CHECKPOINT

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The sum of the coefficients has one prime number, so option E is incorrect.

According to the Jahn-Teller effect, the electronic configuration of Os(VI) is $t_{2g}^2 e_g^0$, which will cause distortion and is not a regular octahedron, so option F is incorrect.

CHECKPOINT

1 PTS

According to the Jahn-Teller effect, the electronic configuration of Os(VI) is $t_{2g}^2 e_g^0$, which will cause distortion and is not a regular octahedron

According to the bond length data of OsO_2F_3 , in this polymer structure, Os exists in two environments: one is connected to five F and one O, and the other is connected to three F and three O. Therefore, the valences of the two Os are Os(VI), Os(VIII) respectively. Obviously, Os(VII) does not have unpaired electrons. The terminal $\text{Os}=\text{O}$ bond of Os(VI) is much shorter than the $\text{Os}-\text{F}$ bond, so it can be regarded as a flattened octahedral field coordination. In the case of two d electrons, the lowest energy d_{xy} is filled, and there are no unpaired electrons. Therefore, the magnetic moment of this compound is almost 0, so option G is correct.

CHECKPOINT

1 PTS

In OsO_2F_3 , the valences of the two Os are Os(VI), Os(VIII)

CHECKPOINT

1 PTS

The terminal $\text{Os}=\text{O}$ bond of Os(VI) is much shorter than the $\text{Os}-\text{F}$ bond, so it can be regarded as a flattened octahedral field coordination

CHECKPOINT

1 PTS

Os(VI) has two d electrons, filling the lowest energy d_{xy} , with no unpaired electrons

CHECKPOINT

1 PTS

The magnetic moment of the compound OsO_2F_3 is almost 0